Hugging Face AI Insurance Claims Platform

A local evidence-review system combining React, Express, MongoDB and FastAPI with real Hugging Face inference. Claim outcomes remain human decisions.

32	6	Local
tests passed	live pipeline stages	validated scope

ARCHITECTURE

Separation of responsibilities

React / TypeScript
Claim screens and human review

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Express / MongoDB
Users, uploads, state and audit

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FastAPI / Models
OCR, classification, NER and vision

WORKFLOW

Review from evidence to decision

01 **Create and upload**

Claims support editable descriptions, required-evidence checklists and protected PDF, DOCX, text and image attachments.

02 **Process with real inference**

FastAPI extracts or recognises text and runs classification, general NER, embeddings and image object detection. Partial failures are retained.

03 **Compare source information**

Labelled fields are checked across sources and financial observations retain their original line context; no payment total is inferred.

04 **Human review and history**

Reviewers record outcomes and notes. Later claim or evidence changes invalidate current analysis while retaining earlier versions.

Evidence and verification

The models help an assessor inspect submitted information. Their outputs are not a fraud finding, damage grade, policy decision or payment calculation.

MODEL STACK

Actual inference components

Classification

valhalla/distilbart-mnli-12-1

General zero-shot document categories; keyword overrides are separately labelled.

Named entities

dslim/bert-base-NER

General entities plus separately marked claim-field extraction rules.

Object detection

facebook/detr-resnet-50

COCO objects and boxes; no damaged-part or severity identification.

Embeddings

all-MiniLM-L6-v2

Normalised embeddings and cosine similarity; not factual agreement.

STRUCTURED COMPARISON

What the source comparison can say

MATCH	Normalised source values agree.
MISMATCH	Comparable source values disagree.
MISSING	The field was not present.
UNVERIFIABLE	Only one source or ambiguous input.

VERIFICATION

What was exercised on the real local stack

Automated checks: 15 Node backend tests, 5 frontend tests and 12 Python tests passed. Final builds, Docker images, protected downloads, role boundaries, logout and duplicate-analysis rejection were checked.

Live synthetic workflow: scanned-PDF OCR extracted fictional identifiers and locations. DETR ran on both a synthetic accident photo (car detected) and a schematic (no detections). Evidence edits, stale analyses, saved reviews and reanalysis were browser-tested.

Deployment boundary. Single-instance local prototype; no insurer integration, production tenant isolation, model calibration, real-claim accuracy benchmark or insurer rule engine. Filesystem and MongoDB changes are not one transaction. Model scores and similarities require human interpretation.